

Gabriel Cabral Melo

Full Stack Software Engineer | React | Angular | Vue | Node.js | TypeScript

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Summary of Qualifications

Experienced **Full Stack Software Engineer** specializing in **Frontend** and **Backend** development with a focus on **React, Angular, Vue, and Node.js** since **2023 (3+ years)**. Skilled in building modern, scalable applications using **TypeScript** and **RxJS**, with both **Monolith** and **Microservices** architectures. Proven track record in **full stack architecture**, **performance optimization**, **accessibility (WCAG)**, **automated testing** (Playwright, Vitest, Testing Library), and **agile methodologies** (Scrum/Jira). Experienced in technical team leadership, **DevOps/CI/CD**, **AWS** cloud infrastructure, and **AI-Assisted Development** (Claude Code, Codex CLI, RAG Pipelines, Ralph Loops).

Experience

Quality Engineering Lead at Grupo UND: 02/2026 – Present

In a new role as the first quality hire, I founded the QE department at Grupo UND, leading the test architecture and performance audits to ensure technical system integrity across their **Nuxt 4 (Vue.js)** and **Laravel 12** stacks.

- Performed deep-dive performance audits on **Nuxt 4 (Vue.js)** and **Laravel 12** stacks, identifying and mitigating critical bottlenecks related to **Big O complexity** and memory leaks in global event listeners.
- Architected and deployed a modern test automation infrastructure using **Playwright** (E2E) and **Vitest** (Unit), integrating **Mutation Testing** to target **80%+ effective code coverage**.
- Accelerated the automation baseline by approximately **60%** leveraging LLMs and agentic programming for complex test case design and execution.
- Authored **30+ pages** of technical QA documentation covering financial business rules, regulatory compliance, and risk-prioritized **Gherkin** test suites.
- Conducted code reviews on critical features and API testing via **AI (PRDs for endpoint validation)**, combining manual, automated, and AI-assisted testing, increasing internal client satisfaction by **30%** and reducing task delivery delays by **40%**.
- Executed manual and API tests on new features, validating critical payment flows, containment logic, and third-party provider integrations prior to releases.

Tech Stack: Vue.js (Nuxt 4), Laravel 12, PHP, TypeScript, Playwright, Vitest, Stryker, Gherkin, REST APIs, AI-Assisted Testing, PRD Generation.

Founder & Principal Full Stack Software Engineer at C&S Softwares: 10/2025 – Present

As founder and principal engineer, I lead the full technical buildout of two products: **Conecta Paraná** — a multi-city urban guide platform built as a monorepo (NestJS + Angular SPA + Flutter) — and **Orçamento Fácil** — a mobile construction budgeting app. I serve as Tech Lead and DevOps for a 6-person team, owning architecture, CI/CD, quality, and delivery.

- Directed the architecture of the Conecta Paraná project, applying **DDD** principles to separate business contexts between the mobile app (Citizen) and web admin panel.
- Architected a full monorepo infrastructure (**NestJS + Angular SPA + Flutter**) with **4 Docker environments**, CI/CD via **GitHub Actions**, and automated ARM64 deployment on Oracle Cloud VPS, reducing new developer onboarding from **days to minutes**.
- Developed **Angular** components with **RxJS** for reactive state management and asynchronous data streams, applying best practices in **performance** and **accessibility (WCAG)**.
- Guided a team of **6 junior developers** as Tech Lead and DevOps, defining architecture, tech stack, conducting code reviews, test reviews, and coordinating sprints via **Jira/Scrum**.
- Enforced end-to-end **CI/CD pipelines** with GitHub Actions (lint, unit tests, e2e, multi-arch builds with QEMU, SSH deploy to GHCR), with branch protection preventing merges without validation.
- Leveraged **Claude Code** for continuous delivery, accelerating features that would take **2–3 weeks** to under **2 days** of development, and assisting with CI/CD configuration.
- Raised test coverage on the Orçamento Fácil project (mobile app) from **60% to 96%** with **Stryker (Mutation Testing)**, ensuring mutant coverage across critical business flows.
- Prototyped interfaces for Orçamento Fácil with **Claude Design**, accelerating the visual validation cycle and reducing UI rework by **70%**.
- Implemented **RAG with AI** pipelines in Docker containers (**bypass mode** with **Ralph Loops**), creating PRDs and developing Orçamento Fácil features autonomously, with continuous review to ensure integrity and consistency with the application.

- Used **Codex CLI** for reviewing AI-developed activities and continuous code review, maintaining quality standards and architectural consistency.

Tech Stack: Angular, NestJS, Flutter/Dart, TypeScript, RxJS, Docker, Nginx, Oracle Cloud ARM, GitHub Actions, CI/CD, Prisma ORM, PostgreSQL, Stryker, Claude Code, Claude Design, Codex CLI, RAG Pipelines, Ralph Loops, React/Capacitor.

Full Stack Software Engineer at Good Software Dev: 12/2024 – 07/2025

As a Full Stack Engineer, I worked across multiple client projects handling backend refactoring, infrastructure cost optimization, real-time systems, and API standardization. Key achievements span revenue recovery, AWS cost reduction, and microservices architecture design.

- Recovered **\$70k annually** in revenue by mitigating contract errors via refactoring cron jobs in **Node.js** and **Axios/Express** integrations, reducing communication failures by **60%** and increasing task reliability by **45%**.
- Reduced **MTTR by 50%** and ensured zero-downtime deployments with **GitHub Actions**, **Docker**, and **Kubernetes**.
- Reduced **AWS** infrastructure costs by **81% (\$3.4k/month)** via **MongoDB** index optimization and automated data cleaning pipelines, enabling instance downsizing from M80 → M40.
- Reduced latency by **50%** with **Redis** caching and complex query optimization in **MongoDB**.
- Built a real-time ticketing system in **React** with **Webhooks**, reducing data update latency by **35%**.
- Developed **Node.js** auth/log middleware to standardize the company's microservices architecture, onboarding **3 new services** within the first month.
- Documented APIs via **OpenAPI/Swagger**, optimizing integration and cutting errors in **half**.
- Built scalable **REST** microservices with **React**, ensuring high system availability.

Tech Stack: Node.js, Express, React, TypeScript, MongoDB, Redis, AWS, Docker, Kubernetes, GitHub Actions, Axios, OpenAPI/Swagger, Webhooks, REST APIs, Microservices.

Front End Developer Instructor at Trybe: 01/2024 – 05/2024

As a Front End Instructor, I mentored cohorts of aspiring developers through JavaScript, TypeScript, and React, focusing on code quality, best practices, and accelerating their path to professional readiness.

- Mentored developers in **JavaScript**, **TypeScript**, and **React**, accelerating technical proficiency by **40%**.
- Conducted code reviews and workshops, elevating code quality for over **50 developers**.
- Optimized materials and support via Slack, reducing ticket resolution time from **40 to 20 minutes**.

Tech Stack: JavaScript, TypeScript, React, HTML, CSS, Git, Slack.

Education

BSc of Systems Analysis and Development (similar to Computer Science): UniCesumar (Brazil). Jul/2026 (2.5 years).

Full-Stack Web Development Course: Trybe (Brazil). Dec/2024 (1 year).

GFT Start #7 - Java: DIO — Digital Innovation One. 2025.

Quality Assurance & Bug Management: Alura. 2024.

Languages: Portuguese (Native), English (Professional/Fluent), Spanish (Conversational).

Technical Skills

Front End: Angular, React, Vue (Nuxt 4), TypeScript, JavaScript (ES6/ES7), RxJS, HTML, CSS, Capacitor, Accessibility (WCAG).

Testing: Playwright, Vitest, Testing Library, Pest PHP, Mutation Testing (Stryker/Infection), Manual Testing, API Testing.

AI-Assisted Development: Claude Code, Claude Design, Codex CLI, RAG Pipelines, Agentic Programming, Ralph Loops, PRD Generation.

Back End & APIs: Node.js, NestJS, Express, REST APIs, Microservices, Axios, OpenAPI/Swagger, Redis, Webhooks.

Databases: PostgreSQL, MongoDB, Firestore, IndexedDB, Prisma ORM.

Cloud & Infrastructure: AWS, Oracle Cloud, Docker, Kubernetes, Nginx, GitHub Actions, CI/CD, Linux.

DevOps & Tools: Git, Jira, Scrum, Docker Compose, GHCR, Certbot, iptables, DDD.